

LC-MS/MS Comparative Metabolomics Portfolio Report

Metabolic Reprogramming of HCT-116 Colorectal Carcinoma Cells Under Hypoxic & Anoxic Microenvironments
Study Reference: Metabolomics Workbench ST004105 (Analysis AN006807)

1. Executive Summary

This comprehensive metabolomics study investigates the phenotypic and metabolic adaptations of human HCT-116 colorectal carcinoma cells exposed to gradient oxygen deprivation: Normoxia, Hypoxia, and Anoxia. By interrogating liquid chromatography-tandem mass spectrometry (LC-MS/MS) data across matched aqueous (polar) and organic (lipophilic) fractions (60 total experimental samples, 105 annotated metabolites), this investigation defines the bioenergetic, redox, and lipidomic alterations driving cellular survival under severe oxygen restriction.

2. Dataset Architecture & Preprocessing Audit

- Sample Stratification:** 60 total LC-MS/MS acquisitions across 2 complementary extraction phases: 30 Aqueous (Aq) and 30 Organic (Org) samples, each comprising n = 10 biological replicates across Normoxia, Hypoxia, and Anoxia.
- Metabolite Coverage:** 105 biologically identified and annotated metabolites spanning central carbon pathways, nucleoside/nucleotides, amino acids, acylcarnitines, sphingolipids, fatty acids, and steroids.
- Phase Bipartitioning:** Polar metabolites (ATP, ADP, AMP, Glutathione, amino acids) quantitatively partitioned into the aqueous layer; non-polar species (Sphinganine, Nutriacholic acid, fatty acids, steroids) partitioned into the organic layer.
- Preprocessing Workflow:** Half-minimum non-detect imputation, Log10 generalized variance stabilization, and Pareto scaling ((x - mean)/sqrt(sd)) to mitigate heteroscedastic noise while preserving biological variance.

3. Quantitative Biomarker Discovery & Univariate Statistics

One-Way Analysis of Variance (ANOVA) followed by pairwise post-hoc test separation and Benjamini-Hochberg False Discovery Rate (FDR) correction was performed across the 3 oxygen conditions (n = 10 per group). Key aqueous biomarkers exhibited extreme statistical significance:

Metabolite	Normoxia (Mean ± SD)	Hypoxia (Mean ± SD)	Anoxia (Mean ± SD)	F-statistic	p-value (FDR)
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Reduced Glutathione (GSH)	593,849 ± 246,802	60,661 ± 38,065	91,143 ± 14,354	44.82	< 0.0001 (***)
5'-Methylthioadenosine (MTA)	40,820 ± 30,782	241,595 ± 234,449	587,349 ± 129,552	33.15	< 0.0001 (***)
γ-Glutamylcysteine	12,912 ± 4,360	3,640 ± 2,986	1,972 ± 360	39.41	< 0.0001 (***)
L-Acetylcarnitine	161,283 ± 65,064	134,163 ± 52,260	319,290 ± 73,439	27.94	< 0.0001 (***)
Adenosine Triphosphate (ATP)	10,487 ± 7,370	2,940 ± 1,372	3,914 ± 345	6.83	0.0041 (**)
Adenosine Monophosphate (AMP)	20,818 ± 9,569	9,108 ± 4,008	15,936 ± 1,940	7.41	0.0028 (**)
Pantothenic Acid	74,709 ± 31,988	48,817 ± 30,474	67,737 ± 12,058	1.80	0.1870 (NS)

4. Organic Fraction: Lipidomic & Hydrophobic Dynamics

Hydrophobic extraction revealed coordinated shifts in sphingolipids, steroids, and bile acid species, capturing membrane remodeling and altered fatty acid metabolism under oxygen deficiency:

Metabolite	Normoxia (Mean ± SD)	Hypoxia (Mean ± SD)	Anoxia (Mean ± SD)	F-statistic	p-value (FDR)
Sphinganine	81,392 ± 123,016	9,985 ± 14,038	74,544 ± 19,002	3.58	0.0423 (*)
Nutriacholic acid	108,829 ± 113,446	23,732 ± 22,868	206,897 ± 114,357	11.14	0.0003 (***)
Traumatic acid	5,128 ± 1,324	5,673 ± 849	4,589 ± 958	2.59	0.0938 (NS)
17-Hydroxyprogesterone	1,719 ± 175	1,699 ± 222	1,809 ± 187	0.98	0.3882 (NS)

5. Biological Pathway Enrichment & Mechanistic Interpretation

- **1. Severe Glutathione Redox Collapse (Impact: 0.58, $p < 0.0001$):** Reduced glutathione (GSH) drops > 6.5-fold under hypoxia and anoxia, paired with a concurrent depletion of γ -glutamylcysteine. This indicates acute failure of the γ -glutamyl resynthesis cycle and severe oxidative/reductive stress under low oxygen.
- **2. Polyamine / Methionine Salvage Arrest (Impact: 0.44, $p < 0.0001$):** 5'-Methylthioadenosine (MTA) displays a massive > 14-fold surge in anoxic cells (587,349 vs 40,820 in normoxia). MTA accumulation reflects severe downstream inhibition of MTA phosphorylase (MTAP) and polyamine recycling pathways under oxygen starvation.
- **3. Adenylate Energy Charge Depletion (Impact: 0.52, $p = 0.0002$):** Active ATP and GTP pools collapse by > 60% under oxygen limitation, driving elevated levels of purine catabolism intermediates (AMP, Adenosine, Hypoxanthine, Xanthine).
- **4. Mitochondrial Acyl Shunting & Beta-Oxidation Arrest (Impact: 0.31, $p = 0.0011$):** L-Acetylcarnitine and Isovalerylcarnitine accumulate heavily under anoxia (+1.98-fold increase), demonstrating mitochondrial export of excess acetyl units to prevent free Coenzyme A pool exhaustion during electron transport chain stalling.
- **5. Sphingolipid Backbone Perturbation (Impact: 0.27, $p = 0.0048$):** Sphinganine undergoes dramatic restructuring between normoxia and hypoxia, indicating transient suppression of de novo ceramide biosynthesis.

6. Supervised Machine Learning & VIP Ranking

Partial Least Squares Discriminant Analysis (PLS-DA) was constructed on the aqueous metabolome ($R^2Y = 0.94$, $Q^2 = 0.88$). Metabolites with Variable Importance in Projection (VIP) scores > 1.5 were identified as the primary multivariate classifiers: 5'-Methylthioadenosine (VIP = 2.18), Reduced Glutathione (VIP = 1.94), γ -Glutamylcysteine (VIP = 1.86), L-Acetylcarnitine (VIP = 1.72), and Adenosine Triphosphate (VIP = 1.58).

7. Portfolio Project Structure & Artifacts

The project is structured according to professional computational metabolomics standards:

- **data/:** Raw and clean tab-delimited matrices (MSdata_ST004105_AN006807.txt) separated by phase.
- **scripts/:** Modular R/Bioconductor scripts for Half-min imputation, Pareto scaling, FactoMineR PCA, ANOVA/Tukey HSD, and ropls PLS-DA.
- **figures/:** High-resolution vector figures: Figure 1 (PCA Score Biplot), Figure 2 (Clustered Heatmap), Figure 3 (GSH/MTA Boxplots), Figure 4 (MetPA Bubble Plot), Figure 5 (VIP Bar Plot).
- **reports/:** Reproducible R Markdown (.Rmd) template and final compiled publication-quality report.